Wazuh Setup and Configuration

Since I already have Docker running, we will deploy the Wazuh single-node stack (Manager, Indexer, and Dashboard) via Docker Compose.

Step 1.Increase Virtual Memory Mapping: Required for Wazuh Indexer.

The Wazuh Indexer (based on OpenSearch) requires a higher virtual memory map count than the Linux default. If skipped, the container will crash immediately.
Code for Debian host:

```
sudo sysctl -w vm.max_map_count=262144
```

To make it persistent across reboots, add it to /etc/sysctl.conf:

```
Bash
echo "vm.max_map_count=262144" | sudo tee -a /etc/sysctl.conf
```

2.Update UFW Firewall Rules:

Because I have a "Default Deny" policy, I must explicitly allow the ports Wazuh requires for agent communication and dashboard access.

```
Bash
# Wazuh Dashboard (Web UI)
sudo ufw allow 443/tcp
# Agent Registration Service
sudo ufw allow 1515/tcp
# Agent Connection Service
sudo ufw allow 1514/tcp
sudo ufw reload
```

3.Clone the Wazuh Docker Repository:

Pull the official Wazuh Docker deployment files. We will use the single node deployment, which is lightweight enough for the Latitude E5440.

```
Bash
git clone https://github.com/wazuh/wazuh-docker.git -b v4.14.6
cd wazuh-docker/single-node
```

4.Generate SSL Certificates:

Wazuh components communicate over TLS. A provided Docker utility automates certificate generation for the stack.

```
Bash
docker compose -f generate-indexer-certs.yml run --rm generator
```

Note: This creates the necessary .pem files in the config/wazuh_indexer_ssl_certs directory.

5.Deploy the Stack:

Launch the Wazuh Manager, Indexer, and Dashboard in the background.

```
Bash
docker compose up -d
```

It takes about 1-2 minutes for the Indexer to fully initialize. You can monitor the progress with docker compose logs -f wazuh.indexer.

6.Access the Dashboard:
Once the stack is running, navigate to `https://<YOUR_IP>` in your browser.

- **Username:** admin
- **Password:** SecretPassword *(This is the default defined in docker-compose.yml — you should change this immediately post-deployment).*

Verifying Resource Usage

Given that this is running alongside Pi-hole on a laptop, keep an eye on hardware utilization. The Wazuh Indexer is Java-based and can be memory-intensive. Run htop to ensure the E5440 has enough overhead remaining. If memory is heavily constrained, you may need to limit the JVM heap size in the docker-compose.yml environment variables before scaling up your agent deployments.